Module 0 Homework 2 Challenge

Provide the best answer for each of the following questions.

Time Remaining 00:59:20

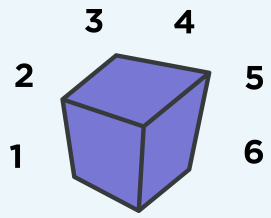
Main Question Set

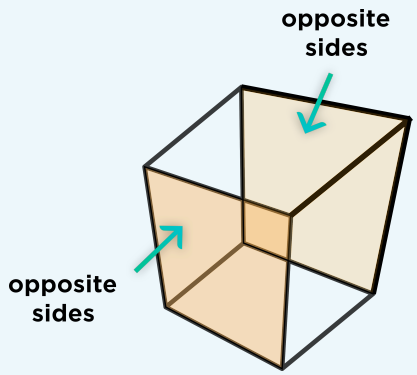
Solve six of the first ten questions to achieve a passing score of 60%. Don't miss the learning opportunity in the rest of the problems, though -- spend the full time and attempt every one of them!

1

Review Later

Standard 6-sided dice with the numbers 1 through 6 have the property that every pair of opposite sides has the same sum. What is that sum?





2

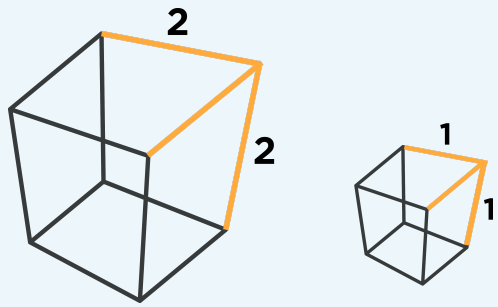
Review Later

How many ways are there to arrange 7 people sitting in a row?

- $7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1$. We have 7 options for the person we place in the first position, 6 options for the second, and so on.
- 7^7. We have 7 people and 7 positions, so we have 7 options per position.
- $7 + 6 + 5 + 4 + 3 + 2 + 1$. We have 7 options for the person we place in the first position, 6 options for the second, and so on.
- $7 + 7 + 7 + 7 + 7 + 7 + 7$. We have 7 people and 7 positions, so we have 7 options per position.
- 7^6. We have 7 people and 7 positions, but after the second to last person is placed, there's only one choice for the last position.

3

Review Later



Suppose you have two cubes of different sizes. The first has a side length of 1 inch and the second has a side length of 2 inches. How many times is the volume of the second cube compared to the volume of the first?

- 2 times as large. The sides are twice as long, so the volume is twice as large.
- 4 times as large. We have to square the side length when calculating volume, so it's $2^2 = 4$ times as large.
- 8 times as large. Volume depends on three dimensions and is calculated by cubing the side length, so it's $2^3 = 8$ times as large.
- 12 times as large. The side length is doubled, but a cube has 6 faces, so it's $2 \times 6 = 12$ times as large.
- Impossible to determine with the given information.

4

Review Later

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100

What is the **average** of the numbers along the diagonal of this square, from the top right to the bottom left?

Your answer should be a decimal, like 1.5

5

Review Later

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100

What is the sum of the numbers along the diagonal of this square, from the top right to the bottom left?

6

Review Later

A multiple of n is a number that can be **divided by** n.

What is the least common multiple of $1, 2, 4, 8, 16, 32, 64, 128$, and 256?

Note: The numbers $1, 2, 4, 8, 16, 32, 64, 128$, and 256 follow a special pattern!

7

Review Later

2		6
?		

This diagram is a partially-filled magic square, where each row, column, and 3-box-long diagonal has the same sum, and the numbers from 1 through 9 each appear exactly once. What number goes in the middle left box?

8

Review Later

Find the value of

$$-9+-8+-7+-6+-5+-4+-3+-2+-1+0+1+2+3+4+5+6+7+8+9+10+11+12.$$

Note: The numbers in red are negative.

9

Review Later

Can we make a magic square with the numbers

$$2, 3, 4, 5, 6, 7, 8, 9, 10$$

?

- No. The sum of the first 10 numbers is 55, which isn't divisible by 3.
- Yes. We can still find ways to make each row sum to 15.
- No. Substituting 10 for 1 destabilizes the sums in a way that can't be fixed.

- Yes. The row sum changes from 15 to 18, but it's possible to make a magic square.
- No. The largest number in a magic square has to be divisble by three.

10

Review Later

If there are 6 people in a room, the number of ways to pick 3 of them is exactly (6 × 5 × 4) / (3 × 2 × 1). Why should we divide by 3 × 2 × 1?

- We have 6 people so we need to do something with all of the numbers 6, 5, 4, 3, 2, and 1. Since we are picking half the people, we multiply the first half together, and divide by the second half.
- The 6 × 5 × 4 corresponds to picking 3 people in order, and we divide by the number of ways we count each combination, which is the number of ways to rearrange the 6 − 3 = 3 people who were not picked.
- The 6 × 5 × 4 corresponds to picking 3 people in order, and we divide by the number of ways we count each combination, which is the number of ways to rearrange the 3 people who were picked.
- We split the 6 people into 3 groups (since 6 = 3 + 2 + 1), and then from the first group we pick 3, from the second group we pick 2, and from the third group we pick 1. That makes 3 × 2 × 1.
- The formula is actually wrong. We should subtract instead of dividing, so the answer should be (6 × 5 × 4) − (3 × 2 × 1).

Bonus Challenge Questions

These optional questions will not count towards your final score. However, we strongly suggest trying them for the benefit of the challenge!

11

Review Later

You're behind a car in the right-turn lane at a traffic light. The car in front has blinkers that flash every 1.2 seconds. Your car's blinker flashes every 1 second. Both cars just flashed simultaneously. How many seconds later will both cars flash simultaneously again?

12

Review Later

What is the least common multiple of $91, 1001$ and 26?

13

Review Later

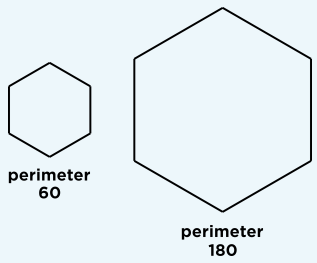
How many ways are there to choose 8 people from a group of 10 people at a big party?

14

Review Later

A regular hexagon (with all sides equal and all angles equal) has perimeter 60. Another regular hexagon has perimeter 180. What is the area of the SMALLER hexagon divided by the area of the LARGER hexagon?

Note: Regular means that all sides are the same length.

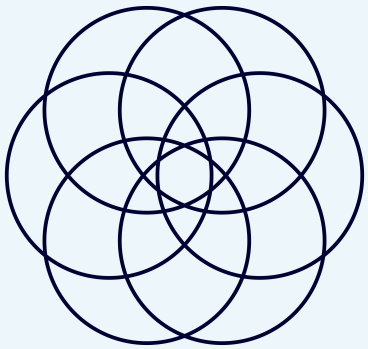


Your answer should be a fraction, written in simplest terms, with no spaces. If the answer were 1/2, you would put 1 in the first blank, and 2 in the second blank.

/

15

Review Later

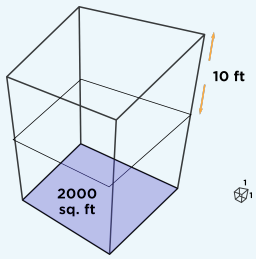


How many intersections are there between the circles in this picture? It may be helpful to count the number of times two specific circles intersect.

16

Review Later

How many 1-centimeter cubes would fit in a 2000-square-foot 2-story house? Suppose the ceiling height of each level is 10 feet. For reference, there are about 30 centimeters in a foot. Choose the closest answer.



- 100
- 1 thousand
- 1 million
- 1 billion

17

Review Later

Suppose that a is a positive integer, and b is a multiple of a. Which of the following are true? Select all answers that apply.

- [ ] b is strictly bigger than a.
- [ ] The prime factorizations of a and b have exactly the same sets of primes in them.
- [ ] Every prime in the prime factorization of a is also in the prime factorization of b.
- [ ] Every prime in the prime factorization of b is also in the prime factorization of a.
- [ ] Each prime in the prime factorization of b has a power that is greater than or equal to its power in the prime factorization of a.

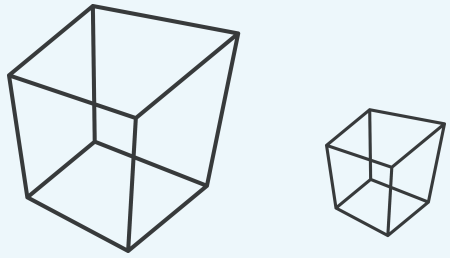
Super Bonus Questions

These questions are an extreme challenge! Some of the most interesting, creative and clever questions are the hardest ones. So try every single one. Even if you can't figure it out, you'll still be deepening your understanding of the important math concepts that underlie this course.

18

Review Later

A factory is painting cubes. If the amount of paint used to cover the entire surface of a big cube is exactly 4 times the amount of paint used to cover a small cube, what is the volume of the big cube divided by the volume of the small cube?



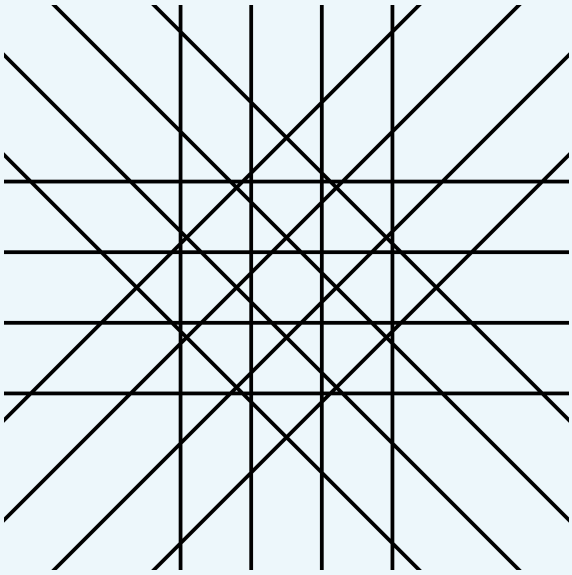
19

Review Later

A pizza store sells 12-inch diameter pizzas and 16-inch diameter pizzas. What percent larger are the 16-inch pizzas, rounded to the nearest percent? (If the answer is 31.4%, you would type in 31.)

20

Review Later



How many triangles are in this picture? A triangle can have lines through it.

View Marked Questions

Finished

Software by
brillium
Version 11.2

Privacy Policy. Assessment content is copyright 2026, Expii, Inc..